

HINTS & SOLUTIONS

Q.1 (D)

$$[H^+]_{\text{resultant}} = \frac{25 \times 0.1 + 25 \times 0.1 \times 2 - 50 \times 0.1}{100}$$

$$= 2.5 \times 10^{-2}$$

$$\text{pH} = 2 - \log(2.5) \Rightarrow 1.6$$

$$\text{pOH} = 14 - 1.6 \Rightarrow 12.6$$

Q.2 (D)

Q.3 (D)

Q.4 (B)

$$\alpha_1 = \sqrt{\frac{k_w}{k_a \cdot C_1}} \quad \frac{\alpha_1}{\alpha_2} = \frac{\sqrt{C_2}}{\sqrt{C_1}}$$

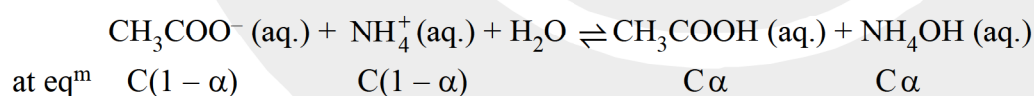
$$\alpha_2 = \sqrt{\frac{k_w}{k_a \cdot C_2}} \quad \frac{0.01}{\alpha_2} = \sqrt{\left(\frac{0.4}{0.1}\right)} = 2$$

$$\alpha_2 = \frac{0.01}{2} = 0.005$$

$$[OH^-] = 0.4 \times 0.005 \Rightarrow 20 \times 10^{-4} = 2 \times 10^{-3} \text{ M}$$

$$[H^+] = \frac{10^{-14}}{2 \times 10^{-3}} = 5 \times 10^{-12} \text{ M}$$

Q.5 (A)



$$K_h = \frac{K_w}{K_a K_b} = \frac{\alpha^2}{(1-\alpha)^2} \Rightarrow \frac{10^{-14}}{(1.8 \times 10^{-5})^2} = \frac{\alpha^2}{(1-\alpha)^2} \Rightarrow \frac{10^{-2}}{1.8} = \frac{\alpha}{1-\alpha}$$

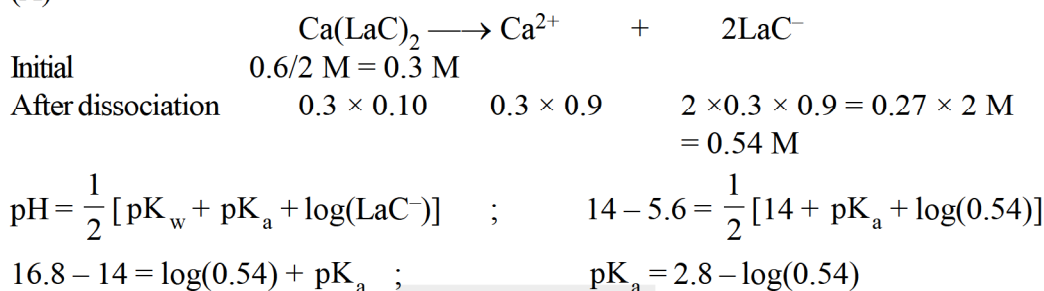
$$[\text{CH}_3\text{COOH}] = C\alpha \Rightarrow 0.15 \times 0.0055 = 8.3 \times 10^{-4} \text{ M}$$

Q.6 (D)

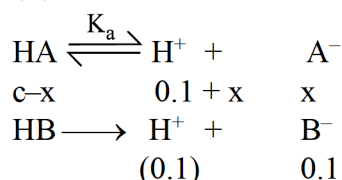
Q.7 (B)

20 meq. of HNO_3 reacts completely with 20 meq of NaOH . Hence, the final solution of neutral. Thus, at 90°C , $\text{pH} = 6$.

Q.8 (A)



Q.9 (A)



$$K_a = \frac{(0.1+x)(x)}{c-x} \quad \{\text{Let } x \ll c \text{ (approx)}\}$$

$$\Rightarrow \text{sub. } 0.1+x=0.1 \quad ; \quad x = \frac{K_a \cdot c}{0.1} = K_a \quad \{\because c=0.1\}$$

$$\alpha = \frac{x}{c} = 10^{-5}$$

Q.10 (D)

Q.11 (D)

$$[\text{H}^+] = 10^{-3} = C\alpha$$

$$K_a = 10^{-5}$$

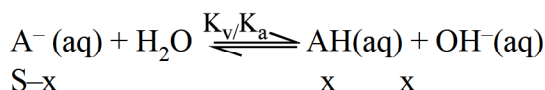
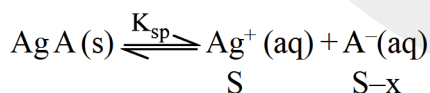
$$K_a = \frac{C\alpha \cdot C\alpha}{C - C\alpha}$$

$$10^{-5} = \frac{10^{-3} \cdot 10^{-3}}{C - 10^{-3}}$$

$$(C - 10^{-3}) = 10^{-1}$$

$$C = 0.101 \text{ M}$$

Q.12 (A)



$$10^{-4} = \frac{K_w}{K_a} = \frac{x^2}{S-x} \Rightarrow \frac{10^{-10}}{S-x} \Rightarrow (S-x) = \frac{10^{-10}}{10^{-4}} = 10^{-6}$$

$$\Rightarrow S = x + 10^{-6} \quad \{\text{but } x = 10^{-5}\} \Rightarrow S = (1.1 \times 10^{-5})$$

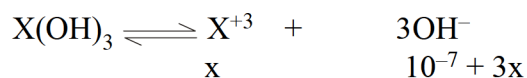
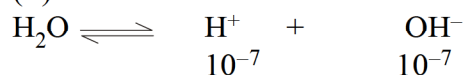
$$K_{sp} = [\text{Ag}^+ \text{(aq)}][\text{A}^- \text{(aq)}] \Rightarrow 1.1 \times 10^{-5} \times 10^{-6}$$

$$K_{sp} = 1.1 \times 10^{-11}$$



Q.13 (C)

Q.14 (B)



$$K_{sp} = [x] [10^{-7} + 3x]^3 \quad 10^{-7} + 3x \approx 10^{-7}$$

$$2.7 \times 10^{-43} = x \times 10^{-21}$$

$$x = 2.7 \times 10^{-22}$$

$$\text{thus } [\text{OH}^-] = 10^{-7} + 3x$$

$$[\text{OH}^-] = 7$$

$$\text{pOH} = 7$$

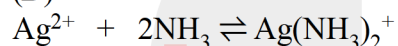
$$\text{pH} = 7$$

Q.15 (D)

Q.16 (A)

Q.17 (A)

Q.18 (D)



$$\frac{0.8-x}{5 \times 10^{-8}} \cdot \frac{z-2x}{z-0.8 \times 2} \cdot \frac{x}{0.8} ; \quad k_f = 10^8; \text{ so } x \approx 0.8$$

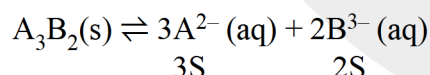
$$k_f = \frac{[\text{Ag(NH}_3)_2^+]}{[\text{Ag}^+][\text{NH}_3]^2} = \frac{0.8}{5 \times 10^{-8} (z-1.6)^2} = 10^8$$

$$(z-1.6)^2 = 0.16; \quad z-1.6 = 0.4 \quad ; \quad z = 2$$

$$\text{Initial concentration } \frac{n}{V} = 2 (\because V = 2 \text{ litre}) ; \quad \therefore n = 4 \text{ litre}$$

Q.19 (C)

$$\text{Solubility } S = x/M$$



$$K_{sp} = (3S)^3(2S)^2 \Rightarrow 108S^5$$

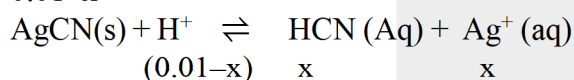
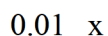
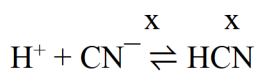
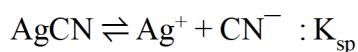
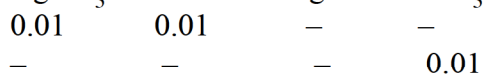
$$\frac{[\text{A}^{3-}]}{K_{sp}} = \frac{2S}{108S^5} \Rightarrow \frac{1}{54S^4} \Rightarrow \frac{1}{54} \frac{M^4}{x^4}$$

Q.20 (D)

pH of any acidic buffer solution may have any value & always tends towards 7 on dilution, hence it may increase or decrease with dilution.



Q.21 (D)



$$10^{-6} = \frac{x^2}{(0.01-x)} = \frac{x^2}{0.01}$$

$$x = 10^{-4}$$

Q.22 (A)

$$[\text{H}^+] = 10^{-2}$$

Q.23 (C)

$$[\text{CN}^-] = \frac{K_{sp}}{[\text{Ag}^+]} = \frac{4 \times 10^{-16}}{10^{-4}} = 4 \times 10^{-12}$$

Q.24 (A)

For AgCl

$$[\text{Ag}^+] = \frac{10^{-10}}{10^{-2}} = 10^{-8}$$

For Ag₂CrO₄

$$[\text{Ag}^+] = \sqrt{\frac{K_{sp}}{(\text{CrO}_4^{2-})}} = \sqrt{\frac{4 \times 10^{-14}}{2 \times 10^{-2}}} = \sqrt{2} \times 10^{-6}$$

Q.25 (B)

$$[\text{Cl}^-]_{\text{left}} = \frac{10^{-10}}{\sqrt{2} \times 10^{-6}} = 0.7 \times 10^{-4} = 7 \times 10^{-5}$$

$$\% \text{Cl}^- \text{ left} = \frac{7 \times 10^{-5}}{10^{-2}} \times 100 = 0.7 \%$$

$$\% \text{Cl}^- \text{ ppt.} = 100 - 0.7 = 99.3 \%$$

Q.26 (A) (B) (C) (D)

(A) For conjugate pairs, $K_a \cdot K_b = K_w$
 $\therefore K_a = K_b = 10^{-7}$

(B) $p_{(HA)}^H = \frac{1}{2}(p^{K_a} - \log C) = \frac{1}{2}(7 - \log 0.1) = 4.0$

(C) $p_{(A^-)}^{OH} = \frac{1}{2}(p^{K_b} - \log C) = \frac{1}{2}(7 - \log 0.1) = 4.0$

(D) $[H^+]_{total} = [H^+]_{HA} + [H^+]_{HCl} \approx 0.01 \text{ M}$
 $\therefore pH = 2.0$

Q.27 (A) P (B) R (C) QS

(A) for $A[OH]_3$

$$k_{sp} = 5 \times 10^{-31} = 4s^3$$

$$s \approx 10^{-10}$$

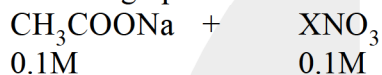
$$[OH^-] \approx 10^{-10} + 10^{-5}$$

(by water)

$$\approx 10^{-5}$$

$$pH = 7$$

(B) On mixing equal volume



In solution

$$[\text{CH}_3\text{COO}^-] = 0.05\text{M}$$

$$[X^+] = 0.05\text{M}$$

$[\text{CH}_3\text{COOX}]$ is salt of weak acid + weak base

$$pH = 4 + \frac{1}{2} pK_a - \frac{1}{2} pK_b$$

$$[\text{for } X^+ \quad K_h = \frac{k_w}{k_b}]$$

$$= 7 + \frac{1}{2} \times 5 - \frac{1}{2} \times 9 = 5$$

(C)

Initial concⁿ



At $\frac{10}{11}$ th equivalence point



$$pOH = pK_b + \log \frac{[\text{Conjugate acid}]}{[\text{Base}]}$$

$$= pK_b + 1 = 10$$

$$pH = 4$$

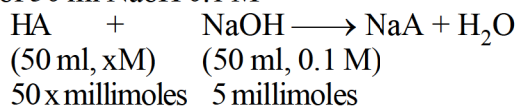
Q.28 0.05 mol



Q.29 0505

∴ after adding of NaOH to weak monoprotic acid, pH is in the acidic medium i.e. NaOH added is quite lesser

After addition of 50 ml NaOH 0.1 M



Left mmols in solution (50x - 5) 0 5

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$4.699 = \text{pK}_a + \log \left(\frac{5}{50x - 5} \right) \dots (i)$$

After adding of 75 ml NaOH 0.1 M

Left millimoles of HA = 50x - 7.5

Millimoles of NaA formed = 7.5

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$5 = \text{pK}_a + \log \left(\frac{7.5}{50x - 7.5} \right) \dots (ii)$$

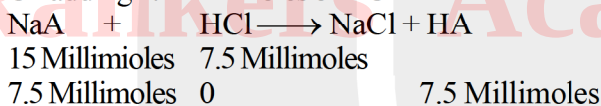
On subtracting (ii) - (i)

$$0.3010 = \log \left(\frac{7.5}{50x - 7.5} \times \frac{(50x - 5)}{5} \right) = \log 2$$

on solving x = 0.3

$$\therefore \text{pK}_a = 5 \Rightarrow ab = 05$$

On adding 7.5 millimoles of HCl



$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]} = \text{pK}_a + \log \frac{7.5}{7.5}$$

$$\text{pH} = \text{pK}_a = 5 \Rightarrow cd = 05$$

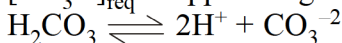
Q.30 4860

$$K_{\text{sp}} \text{MgCO}_3 = 2.5 \times 10^{-7} = [\text{Mg}^{+2}][\text{CO}_3^{-2}]$$

$$K_{\text{sp}} \text{SrCO}_3 = 9 \times 10^{-9} = [\text{Sr}^{+2}][\text{CO}_3^{-2}]$$

$$[\text{CO}_3^{-2}]_{\text{req}} \text{ for ppt of SrCO}_3 = 9 \times 10^{-8}$$

$$[\text{CO}_3^{-2}]_{\text{req}} \text{ for ppt of MgCO}_3 = 2.5 \times 10^{-5}$$



$$K_{\text{a overall}} = \frac{[\text{H}^+][\text{CO}_3^{-2}]}{[\text{H}_2\text{CO}_3]}$$

$$[\text{H}^+]^2 = \frac{5 \times 10^{-16} \times 0.05}{9 \times 10^{-8}} \quad \text{when Sr}^{+2} \text{ will ppt.}$$

$$[\text{H}^+]^2 = \frac{5 \times 10^{-16} \times 0.05}{2.5 \times 10^{-5}} \quad \text{when Mg}^{+2} \text{ will ppt.}$$

$$\text{pH}_1 = 4.8$$

$$\text{pH}_2 = 6.0 \quad = 4860 \text{ Ans.}$$

